

GATE DA Practice Set 3 — Descriptive Statistics, Decision Trees, Permutation & Combination

Coverage: Descriptive Statistics (Mean/Median/Mode/SD); Decision Trees (Entropy, Information Gain, Gini); Permutation & Combination basics.

Sourcing note: Same honesty convention as Sets 1-2. **PYQ-PATTERN** = mirrors a well-known real GATE-style question structure/numbers but is not a verified exact paper question; **ORIGINAL** = newly written, GATE DA-level. No fabricated years or question numbers are used.

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1 1. Descriptive Statistics (Mean/Median/Mode/SD)

Q1 NAT PYQ-PATTERN EASY

Find the median of: 3, 5, 7, 9, 11.

ANSWER7

SOLUTION

Data is already sorted with 5 values (odd count); the median is the middle value = 7.

Concept: Median of an odd-sized dataset

Q2 NAT PYQ-PATTERN MEDIUM

Find the median of: 4, 8, 6, 5, 3, 2.

ANSWER4.5

SOLUTION

Sorted: 2,3,4,5,6,8 (n=6, even). Median = average of 3rd and 4th values = $(4+5)/2 = 4.5$.

Concept: Median of an even-sized dataset

Q3 NAT PYQ-PATTERN EASY

Find the mode of: 2, 3, 3, 4, 4, 4, 5.

ANSWER4

SOLUTION

4 occurs 3 times, more than any other value, so mode = 4.

Concept: Mode identification

Q4 MCQ PYQ-PATTERN EASY

Which measure of central tendency is most sensitive to extreme outliers?

- (A) Mean
- (B) Median
- (C) Mode
- (D) All are equally sensitive

ANSWER(A)

SOLUTION

The mean uses every value's magnitude in its calculation, so a single extreme outlier can shift it substantially; median and mode are comparatively robust.

Concept: Robustness of central tendency measures

Q5 NAT PYQ-PATTERN MEDIUM

Find the (population) variance of: 2, 4, 6, 8, 10.

ANSWER8

SOLUTION

Mean = 6. Deviations: $-4, -2, 0, 2, 4$. Squared: 16,4,0,4,16, sum=40. Variance = $40/5 = 8$.

Concept: Population variance calculation

Q6 NAT PYQ-PATTERN MEDIUM

Using Q6 (variance = 8), find the standard deviation (round to 4 decimals).

ANSWER2.8284

SOLUTION

$SD = \sqrt{\text{Variance}} = \sqrt{8} \approx 2.8284$.

Concept: Standard deviation from variance

Q7 MSQ PYQ-PATTERN MEDIUM

If a constant c is added to every value in a dataset, which of the following are TRUE?

- (A) The mean increases by c
- (B) The standard deviation is unchanged
- (C) The variance is unchanged
- (D) The median increases by c

ANSWER(A), (B), (C), (D)

SOLUTION

Adding a constant shifts every value equally: mean and median shift by c , but spread measures (variance, SD) are unaffected since relative distances between points don't change.

Concept: Effect of a linear shift on summary statistics

Q8 MCQ PYQ-PATTERN MEDIUM

If every value in a dataset is multiplied by a constant k , the standard deviation:

- (A) Remains unchanged
- (B) Gets multiplied by k^2
- (C) Gets multiplied by $|k|$
- (D) Gets divided by k

ANSWER(C)

SOLUTION

Scaling every value by k scales the SD by $|k|$ (variance scales by k^2 , and SD is its square root).

Concept: Effect of scaling on standard deviation

Q9 NAT PYQ-PATTERN MEDIUM

Mean = 50, SD = 5. Find the coefficient of variation ($CV = SD/\text{mean} \times 100$), in %.

ANSWER10

SOLUTION

$CV = (5/50) \times 100 = 10\%$.

Concept: Coefficient of variation

Q10 NAT ORIGINAL EASY

Find the mean of: 10, 20, 30, 40, 50.

ANSWER30

SOLUTION

Sum = 150, $n = 5$. Mean = $150/5 = 30$.

Concept: Arithmetic mean

Q11 NAT ORIGINAL EASY

Find the mode of: 1, 2, 2, 3, 3, 3, 4.

ANSWER3**SOLUTION**

3 appears 3 times, the highest frequency, so mode = 3.

Concept: Mode identification

Q12 NAT ORIGINAL MEDIUM

Find the median of: 12, 15, 11, 18, 10, 20, 14.

ANSWER14**SOLUTION**

Sorted: 10,11,12,14,15,18,20 (n=7, odd). Median = middle (4th) value = 14.

Concept: Median with sorting required first

Q13 NAT ORIGINAL MEDIUM

Find the (population) variance of: 5, 7, 9, 11, 13.

ANSWER8**SOLUTION**

Mean = 9. Deviations: $-4, -2, 0, 2, 4$. Squared sum = $16+4+0+4+16 = 40$. Variance = $40/5 = 8$.

Concept: Population variance

Q14 NAT ORIGINAL MEDIUM

Using Q14 (variance = 8), find the standard deviation (round to 4 decimals).

ANSWER2.8284**SOLUTION**

SD = $\sqrt{8} \approx 2.8284$.

Concept: Standard deviation from variance

Q15 MCQ ORIGINAL MEDIUM

For a right-skewed (positively skewed) distribution, which ordering of central tendency measures typically holds?

- (A) Mean < Median < Mode
- (B) Mean = Median = Mode
- (C) Mean > Median > Mode
- (D) Mode > Mean > Median

ANSWER(C)**SOLUTION**

A long right tail pulls the mean upward more than the median, and the mode (peak of the bulk of data) is typically the smallest of the three: Mean > Median > Mode.

Concept: Skewness and relative position of central tendency measures

Q16 NAT ORIGINAL MEDIUM

Find the sample variance (using $n-1$ in the denominator) of: 4, 8, 6, 10, 7.

ANSWER5**SOLUTION**

Mean = $35/5 = 7$. Deviations: $-3, 1, -1, 3, 0$. Squared: $9, 1, 1, 9, 0$, sum=20. Sample variance = $20/(5-1) = 5$.

Concept: Sample variance (Bessel's correction)

Q17 NAT ORIGINAL HARD

Using Q17 (sample variance = 5), find the sample standard deviation (round to 4 decimals).

ANSWER2.2361

SOLUTION
Sample SD = $\sqrt{5} \approx 2.2361$.

Concept: Sample standard deviation

Q18 NAT ORIGINAL HARD

Dataset A has $n_1=4$, $\text{mean}_1=10$. Dataset B has $n_2=6$, $\text{mean}_2=20$. Find the combined mean of A and B together.

ANSWER16

SOLUTION
Combined mean = $(n_1 \cdot \text{mean}_1 + n_2 \cdot \text{mean}_2) / (n_1 + n_2) = (4 \times 10 + 6 \times 20) / 10 = (40 + 120) / 10 = 16$.

Concept: Combined (pooled) mean of two datasets

Q19 NAT ORIGINAL HARD

Data: 3, 6, 9, 12, 15 has mean 9 and standard deviation $\sqrt{18} \approx 4.2426$. Each value is transformed as $y = 2x + 5$. Find the new standard deviation (round to 4 decimals).

ANSWER8.4853

SOLUTION
Additive constants don't affect SD; multiplicative constants scale it by their absolute value. New SD = $2 \times 4.2426 \approx 8.4853$.

Concept: Effect of a linear transformation on standard deviation

Answer Key — Descriptive Statistics

1-5.7 | 2-7 | 3-4.5 | 4-4 | 5-A | 6-8 | 7-2.8284 | 8-ABCD | 9-C | 10-10 | 11-30 | 12-3 | 13-14 | 14-8 | 15-2.8284 | 16-C | 17-5 | 18-2.2361 | 19-16 | 20-8.4853

2. Decision Trees — Entropy, Information Gain, Gini

Q1 NAT PYQ-PATTERN MEDIUM

Find the entropy (in bits) for a binary split with proportions $p=0.9, 0.1$ (round to 4 decimals).

ANSWER 0.4690

SOLUTION

Entropy = $-0.9 \cdot \log_2(0.9) - 0.1 \cdot \log_2(0.1) = 0.9(0.1520) + 0.1(3.3219) = 0.1368 + 0.3322 = 0.4690$.

Concept: Entropy calculation for skewed proportions

Q2 MCQ PYQ-PATTERN EASY

For a binary classification node, entropy is maximized when:

- (A) One class dominates completely ($p=1, 0$)
- (B) The two classes are equally likely ($p=0.5, 0.5$)
- (C) The node is pure
- (D) There are more than 2 classes

ANSWER (B)

SOLUTION

Entropy is a measure of uncertainty and peaks when outcomes are equally likely — for a binary split, at $p=0.5$.

Concept: Conditions for maximum entropy

Q3 NAT PYQ-PATTERN EASY

Find the Gini index for a binary split with $p=0.5, 0.5$.

ANSWER 0.5

SOLUTION

Gini = $1 - (p_1^2 + p_2^2) = 1 - (0.25 + 0.25) = 0.5$.

Concept: Gini index formula, maximum-impurity case

Q4 NAT PYQ-PATTERN MEDIUM

Find the Gini index for a binary split with $p=0.8, 0.2$.

ANSWER 0.32

SOLUTION

Gini = $1 - (0.8^2 + 0.2^2) = 1 - (0.64 + 0.04) = 1 - 0.68 = 0.32$.

Concept: Gini index calculation

Q5 MCQ PYQ-PATTERN EASY

For binary classification, the Gini index ranges over:

- (A) 0 to 1, maximum at $p=0.5$
- (B) 0 to 0.5, maximum at $p=0.5$
- (C) -1 to 1
- (D) 0 to 0.5, maximum at $p=0$ or 1

ANSWER (B)

SOLUTION

For two classes, Gini = $1 - (p^2 + (1-p)^2) = 2p(1-p)$, which ranges from 0 (pure) to a maximum of 0.5 at $p=0.5$.

Concept: Range and extremes of the Gini index

Q6 NAT PYQ-PATTERN MEDIUM

A dataset has 14 examples: 9 positive, 5 negative. Find the entropy of this set (round to 4 decimals).

ANSWER0.9402

SOLUTION

$p_+ = 9/14 \approx 0.6429$, $p_- = 5/14 \approx 0.3571$.

Entropy = $-0.6429 \cdot \log_2(0.6429) - 0.3571 \cdot \log_2(0.3571) = 0.6429(0.6374) + 0.3571(1.485) \approx 0.4098 + 0.5304 = 0.9402$.

Concept: Entropy from class counts

Q7 MSQ PYQ-PATTERN MEDIUM

Which of the following are TRUE about entropy and Gini index in decision trees?

- (A) Entropy = 0 for a pure node
- (B) Gini index = 0 for a pure node
- (C) A higher information gain is generally preferred when choosing a split
- (D) Entropy and Gini always produce identical tree structures on any dataset

ANSWER(A), (B), (C)

SOLUTION

Entropy and Gini are different impurity measures and can occasionally favor different splits, so (D) is false. (A), (B), (C) are standard, correct facts.

Concept: Impurity-measure fundamentals and common misconception

Q8 NAT PYQ-PATTERN MEDIUM

Parent entropy = 1. A split creates two equal-sized child nodes, each with entropy 0. Find the information gain.

ANSWER1

SOLUTION

IG = Entropy(parent) – weighted entropy(children) = $1 - (0.5 \times 0 + 0.5 \times 0) = 1$.

Concept: Information gain with pure children

Q9 NAT PYQ-PATTERN HARD

A parent node has 10 examples: 6 "yes", 4 "no" (entropy ≈ 0.9710). A split creates: Child1 (5 examples: 4 yes, 1 no; entropy ≈ 0.7219) and Child2 (5 examples: 2 yes, 3 no; entropy ≈ 0.9710). Find the information gain (round to 4 decimals).

ANSWER0.1245

SOLUTION

Weighted child entropy = $0.5(0.7219) + 0.5(0.9710) = 0.3610 + 0.4855 = 0.8465$.

IG = $0.9710 - 0.8465 = 0.1245$.

Concept: Full information gain calculation with a weighted split

Q10 NAT ORIGINAL EASY

Find the entropy of a node where $p=1, 0$ (a pure node).

ANSWER0

SOLUTION

A pure node has zero uncertainty: Entropy = $-1 \cdot \log_2(1) - 0 \cdot \log_2(0) = 0$ (by convention, $0 \cdot \log_2(0) = 0$).

Concept: Entropy of a pure node

Q11 MCQ ORIGINAL EASY

When choosing the best attribute to split on in a decision tree using information gain, we select the attribute that:

- (A) Minimizes information gain
- (B) Maximizes information gain (i.e., reduces impurity the most)
- (C) Has the most distinct values
- (D) Has the fewest training examples

ANSWER(B)**SOLUTION**

The split with the highest information gain reduces uncertainty the most and is preferred.

Concept: Split-selection criterion using information gain

Q12 NAT ORIGINAL MEDIUM

Find the Gini index for a binary split with $p=0.6, 0.4$.

ANSWER0.48**SOLUTION**

$\text{Gini} = 1 - (0.36 + 0.16) = 1 - 0.52 = 0.48$.

Concept: Gini index calculation

Q13 NAT ORIGINAL MEDIUM

Find the entropy (in bits) for a 3-class node with proportions $p=(0.5, 0.3, 0.2)$ (round to 4 decimals).

ANSWER1.4855**SOLUTION**

$\text{Entropy} = -0.5 \cdot \log_2(0.5) - 0.3 \cdot \log_2(0.3) - 0.2 \cdot \log_2(0.2) = 0.5(1) + 0.3(1.737) + 0.2(2.3219) \approx 0.5 + 0.5211 + 0.4644 = 1.4855$.

Concept: Entropy for a multi-class node

Q14 NAT ORIGINAL MEDIUM

Find the Gini index for a 3-class node with proportions $p=(0.5, 0.3, 0.2)$.

ANSWER0.62**SOLUTION**

$\text{Gini} = 1 - (0.25 + 0.09 + 0.04) = 1 - 0.38 = 0.62$.

Concept: Gini index for a multi-class node

Q15 MCQ ORIGINAL MEDIUM

Compared to entropy, the Gini index is generally preferred in some implementations (e.g., CART) mainly because:

- (A) It always gives a higher value than entropy
- (B) It avoids computing logarithms, making it computationally cheaper
- (C) It can only be used for regression trees
- (D) It ranges from -1 to 1

ANSWER(B)**SOLUTION**

Gini uses only squares and sums (no logarithm), making it faster to compute than entropy while giving qualitatively similar splits in practice.

Concept: Practical comparison of Gini vs. entropy

Q16 MSQ ORIGINAL MEDIUM

Which of the following are TRUE about decision trees?

- (A) A decision "stump" (root plus two leaves) has depth 1
- (B) A good split generally decreases (or leaves unchanged) the weighted impurity compared to the parent
- (C) Pruning is used to help reduce overfitting
- (D) Deeper trees always generalize better to unseen data

ANSWER(A), (B), (C)

SOLUTION
Deeper trees tend to overfit rather than always generalize better, so (D) is false. (A), (B), (C) are standard, correct facts about decision trees.
Concept: General decision tree properties and overfitting

Q17 NAT ORIGINAL HARD

A parent node has 20 examples: 12 positive, 8 negative (entropy ≈ 0.9710). A split creates: Child1 (10 examples: 9 pos, 1 neg; entropy ≈ 0.4690) and Child2 (10 examples: 3 pos, 7 neg; entropy ≈ 0.8813). Find the information gain (round to 4 decimals).

ANSWER0.2958

SOLUTION
Weighted child entropy = $0.5(0.4690) + 0.5(0.8813) = 0.2345 + 0.4407 = 0.6752$.
 $IG = 0.9710 - 0.6752 = 0.2958$.
Concept: Multi-step information gain calculation

Q18 NAT ORIGINAL HARD

Using the same split as Q18 (parent Gini=0.48, Child1: 9 pos/1 neg, Child2: 3 pos/7 neg), find the weighted Gini index of the children (round to 2 decimals).

ANSWER0.30

SOLUTION
Child1 Gini = $1 - (0.81 + 0.01) = 0.18$. Child2 Gini = $1 - (0.09 + 0.49) = 0.42$.
Weighted Gini = $0.5(0.18) + 0.5(0.42) = 0.09 + 0.21 = 0.30$.
Concept: Weighted Gini index across a split

Q19 MCQ ORIGINAL HARD

For binary classification, which statement correctly compares entropy and Gini index?

- (A) Both are maximized at $p=0.5$ and minimized (=0) at $p=0$ or $p=1$
- (B) Gini's maximum value is 1, same as entropy's maximum
- (C) Entropy is always smaller than Gini for the same p
- (D) Gini index can be negative for some p

ANSWER(A)

SOLUTION
Both measures peak at maximum uncertainty ($p=0.5$) and hit 0 at a pure node — though their maximum *values* differ (entropy's max is 1 bit, Gini's max is 0.5), the location of the extremes matches, making (A) correct. (B), (C), (D) are false.
Concept: Comparing entropy and Gini behavior — conceptual trap

Answer Key — Decision Trees

1-1 | 2-0.4690 | 3-B | 4-0.5 | 5-0.32 | 6-B | 7-0.9402 | 8-ABC | 9-1 | 10-0.1245 | 11-0 | 12-B | 13-0.48 | 14-1.4855 | 15-0.62 | 16-B | 17-ABC | 18-0.2958 | 19-0.30 | 20-A

3. Permutation & Combination Basics

Q1 NAT PYQ-PATTERN EASY

Find 6C_2 (combinations of 6 items taken 2 at a time).

ANSWER15

SOLUTION

${}^nC_r = n!/(r!(n-r)!)$. ${}^6C_2 = 6!/(2!4!) = 720/48 = 15$.

Concept: Basic combination formula

Q2 NAT PYQ-PATTERN MEDIUM

Find the number of distinct arrangements of the letters of the word "LEVEL".

ANSWER30

SOLUTION

LEVEL has 5 letters with L repeated twice and E repeated twice. Arrangements = $5!/(2! \times 2!) = 120/4 = 30$.

Concept: Permutations with repeated elements

Q3 NAT PYQ-PATTERN EASY

In how many ways can a committee of 3 be selected from 8 people?

ANSWER56

SOLUTION

${}^8C_3 = 8!/(3!5!) = 336/6 = 56$.

Concept: Basic combination application

Q4 NAT PYQ-PATTERN EASY

In how many ways can 4 distinct books be arranged on a shelf?

ANSWER24

SOLUTION

Arranging n distinct items = $n!$. $4! = 24$.

Concept: Basic factorial arrangement

Q5 MCQ PYQ-PATTERN EASY

Which identity is always TRUE for combinations?

- (A) ${}^nC_r = {}^nC_{(n-r)}$
- (B) ${}^nC_r = {}^nC_{r+1}$
- (C) ${}^nC_r = {}^rC_n$
- (D) ${}^nC_r = n \times {}^{n-1}C_r$

ANSWER(A)

SOLUTION

Choosing r items out of n is equivalent to choosing which $(n-r)$ items to leave out, so ${}^nC_r = {}^nC_{(n-r)}$ always holds.

Concept: Symmetry identity of combinations

Q6 NAT PYQ-PATTERN MEDIUM

How many 3-digit numbers can be formed using the digits 1 to 5, without repetition?

ANSWER60**SOLUTION**

This is ${}^5P_3 = 5!/2! = 120/2 = 60$.

Concept: Permutations applied to digit arrangement

Q7 NAT PYQ-PATTERN MEDIUM

A committee is to be formed with 2 men from 5 men and 3 women from 6 women. Find the number of ways.

ANSWER200**SOLUTION**

${}^5C_2 \times {}^6C_3 = 10 \times 20 = 200$.

Concept: Combined selection using the multiplication principle

Q8 MSQ PYQ-PATTERN MEDIUM

Which of the following identities are TRUE?

- (A) ${}^nP_r = n!/(n-r)!$
- (B) ${}^nC_r = n!/(r!(n-r)!)$
- (C) ${}^nC_r = {}^nP_r/r!$
- (D) ${}^nP_r = {}^nC_r + r$

ANSWER(A), (B), (C)**SOLUTION**

(D) is false — the correct relation is ${}^nP_r = {}^nC_r \times r!$ (permutations count each combination in every possible order). (A), (B), (C) are correct standard identities.

Concept: Relationship between permutations and combinations

Q9 NAT PYQ-PATTERN MEDIUM

Find the number of diagonals in a hexagon (6 sides).

ANSWER9**SOLUTION**

Diagonals = ${}^nC_2 - n = 15 - 6 = 9$ (total pairs of vertices minus the n sides themselves).

Concept: Combinations applied to a geometry counting problem

Q10 NAT ORIGINAL EASY

Find 4P_2 .

ANSWER12**SOLUTION**

${}^4P_2 = 4!/2! = 24/2 = 12$.

Concept: Basic permutation calculation

Q11 NAT ORIGINAL EASY

Find 5C_3 .

ANSWER10**SOLUTION**

${}^5C_3 = 5!/(3!2!) = 120/12 = 10$.

Concept: Basic combination calculation

Q12 NAT ORIGINAL MEDIUM

Find the number of distinct arrangements of the letters of the word "STATISTICS" (S=3, T=3, A=1, I=2, C=1).

ANSWER50400**SOLUTION**

Total letters = 10. Arrangements = $10!/(3!\times 3!\times 1!\times 2!\times 1!) = 3628800/(6\times 6\times 1\times 2\times 1) = 3628800/72 = 50400$.

Concept: Permutations with multiple repeated letters

Q13 NAT ORIGINAL MEDIUM

In how many ways can 4 cards be selected from a standard deck of 52 cards?

ANSWER270725**SOLUTION**

${}^{52}C_4 = 52!/(4!\times 48!) = 270725$.

Concept: Combinations with larger numbers

Q14 NAT ORIGINAL MEDIUM

In how many ways can 5 people be seated around a circular table?

ANSWER24**SOLUTION**

Circular permutations of n distinct items = $(n-1)!$. $(5-1)! = 4! = 24$.

Concept: Circular permutation formula

Q15 MCQ ORIGINAL MEDIUM

If repetition is allowed, the number of ways to arrange r items chosen from n types is:

- (A) nP_r
- (B) nC_r
- (C) n^r
- (D) r^n

ANSWER(C)**SOLUTION**

With repetition allowed, each of the r positions independently has n choices, giving n^r total arrangements.

Concept: Permutations with repetition

Q16 NAT ORIGINAL MEDIUM

How many 4-letter "words" (with repetition allowed) can be formed from the 26 letters of the alphabet?

ANSWER456976**SOLUTION**

Each of the 4 positions has 26 choices independently: $26^4 = 456976$.

Concept: Counting with repetition allowed

Q17 NAT ORIGINAL HARD

In how many ways can 10 identical candies be distributed among 3 children so that each child gets at least 1 candy?

ANSWER36**SOLUTION**

Using stars and bars with the "at least 1 each" constraint: distribute $10-3=7$ remaining candies freely among 3 children, i.e., $C(7+3-1, 3-1) = C(9,2) = 36$.

Concept: Stars and bars (distribution counting)

Q18

NAT

ORIGINAL

HARD

In how many ways can 3 men and 3 women be arranged in a row such that men and women alternate?

ANSWER72

SOLUTION

Two alternating patterns are possible (MWMWMW or WMWMWM). For each pattern, the 3 men can be arranged in 3! ways and the 3 women in 3! ways: Total = $2 \times 3! \times 3! = 2 \times 6 \times 6 = 72$.

Concept: Arrangement with an alternating constraint

Q19

NAT

ORIGINAL

HARD

A team of 5 is to be selected from 6 boys and 4 girls such that at least 2 girls are included. Find the number of ways.

ANSWER186

SOLUTION

Case girls=2: ${}^4C_2 \times {}^6C_3 = 6 \times 20 = 120$.
Case girls=3: ${}^4C_3 \times {}^6C_2 = 4 \times 15 = 60$.
Case girls=4: ${}^4C_4 \times {}^6C_1 = 1 \times 6 = 6$.
Total = $120 + 60 + 6 = 186$.

Concept: Case-based combination counting with a minimum constraint

Answer Key — Permutation & Combination

1-20 | 2-15 | 3-30 | 4-56 | 5-24 | 6-A | 7-60 | 8-200 | 9-ABC | 10-9 | 11-12 | 12-10 | 13-50400 | 14-270725 | 15-24 | 16-C | 17-456976 | 18-36 | 19-72 | 20-186

COMPLETE ANSWER KEY — ALL 60 QUESTIONS

Descriptive Statistics: 1-5.7, 2-7, 3-4.5, 4-4, 5-A, 6-8, 7-2.8284, 8-ABCD, 9-C, 10-10, 11-30, 12-3, 13-14, 14-8, 15-2.8284, 16-C, 17-5, 18-2.2361, 19-16, 20-8.4853

Decision Trees: 1-1, 2-0.4690, 3-B, 4-0.5, 5-0.32, 6-B, 7-0.9402, 8-ABC, 9-1, 10-0.1245, 11-0, 12-B, 13-0.48, 14-1.4855, 15-0.62, 16-B, 17-ABC, 18-0.2958, 19-0.30, 20-A

Permutation & Combination: 1-20, 2-15, 3-30, 4-56, 5-24, 6-A, 7-60, 8-200, 9-ABC, 10-9, 11-12, 12-10, 13-50400, 14-270725, 15-24, 16-C, 17-456976, 18-36, 19-72, 20-186